PRAJJAWAL SINGH

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BACKEND ENGINEER | JAVA & SPRING BOOT

Backend Engineer with 1+ year of experience building production APIs, real-time data pipelines and cloud-hosted services. Skills in Java 17+, Spring Boot, Python, SQL, distributed systems and performance optimisation. Reduced customer load times by 60% and monthly infrastructure spending by 38%.

TECHNICAL SKILLS

Languages: Java 17+, Python, C++, SQL, JavaScript

Java: Spring Boot, Spring MVC, Hibernate/JPA, MapStruct, Liquibase/Flyway, JUnit; REST APIs, OAuth2/JWT, Django, FastAPI

Data: PostgreSQL, MySQL, Redis, ClickHouse, ScyllaDB, MongoDB; Kafka, WarpStream, SQS, RabbitMQ; CQRS, Event Sourcing

Cloud & tools: AWS (EKS, RDS, S3, Secrets Manager), CDN, WAF, Docker, Kubernetes, Helm, Ingress, Nginx, GitHub Actions

Engineering: Webhooks, idempotency, retries, HMAC, OpenTelemetry, SigNoz, Grafana, Git, Linux, DSA, OOP, concurrency, system design

EXPERIENCE

Backend Engineer - TailNode | Delhi, India | Jul 2025 - Present

• Investigated slow shape and coordinate loading and rising infrastructure costs by examining content delivery, coordinate payloads, S3 storage usage, and Kafka replication traffic across Availability Zones.

• Implemented CDN caching for frequently accessed content, reducing average load time from 5 seconds to 2 seconds (60%). Monthly origin data-transfer costs fell from $500 to $300 (40%). Integrated a WAF to filter malicious requests and protect application endpoints.

• Removed redundant coordinates and optimised precision while preserving required shape accuracy. Average payload size fell from 2 MB to 800 KB (60%), making shapes faster to download and render and smoother to interact with.

• Optimised stored data and applied lifecycle policies where appropriate, reducing monthly S3 storage costs from $300 to $210 (30%).

• Moved a suitable messaging workload from Kafka to WarpStream to reduce cross-Availability Zone replication traffic. Monthly messaging infrastructure costs fell from $700 to $420 (40%). Across origin transfer, S3 storage, and messaging, monthly spend fell from $1,500 to $930, saving $570 (38%).

• Automated Kafka-to-ClickHouse ETL for 500K+ daily events, cutting reporting latency from hours to under 2 minutes, with replication, CQRS and Event Sourcing for audit-ready data.

• Built secured REST and FastAPI endpoints at sub-50 ms p99 using OAuth2/JWT. Containerized services with Docker and deployed them on Kubernetes using Helm and Ingress, supporting a 99.9% uptime SLA. Configured Nginx and load balancers and debugged production issues across logs, metrics and databases.

• Led design reviews on PostgreSQL sharding and indexing, Redis caching, ScyllaDB write optimisation and distributed consistency. Used Claude Code, Cursor and Copilot with critical code review to ship tested changes.

PROJECTS

Skyline - Social Media Application | Node.js, React, MongoDB
Built authentication, posts and feeds; compound indexes reduced average feed query time by 70%. Implemented GitHub Actions CI/CD and applied MVC and Observer patterns.

Quiz Competition Platform | Redis, Django, React
Delivered real-time leaderboard updates via Redis pub/sub with under 100ms latency for 500+ concurrent participants; reduced page load by 45% through lazy loading, code splitting and query optimisation.

EDUCATION & ACHIEVEMENTS

NIT Tiruchirappalli | B.Tech, Electronics & Communication Engineering | 2021 - 2025 | CGPA: 7.91/10

LeetCode Guardian: 2130 (top 1.92%); CodeChef 4-star (1922 peak); Codeforces Specialist (~1550). Solved 1,000+ DSA problems.